Inferential Statistics & Central Limit Theorem

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Branch of statistics that focuses on drawing conclusions about a population based on a sample.

Population → Entire group of elements

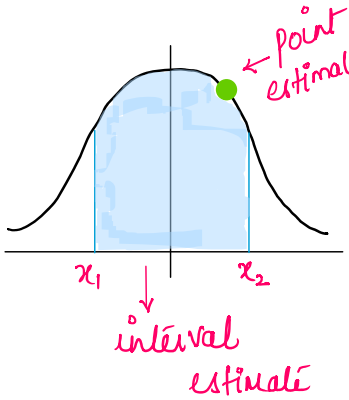
Sample → A subset of population

Ex- Studying the opinion of 10000 people to understand the sentiment of the entire city.

* Limitations of Descriptive Statistics :-
  - Limited Scope → Cannot predict outside the dataset
  - No Generalization → Cannot tell about a population based on a sample
  - No Hypothesis Testing

All the above limitations of Descriptive Statistics can be solved using inferential statistics.

* Concepts used in Inferential Statistics :-
  1. Population
  2. Sample
  3. Parameters → population mean, population std., etc.
  4. Statistics → sample mean, sample std. etc.
  5. Estimation → a) Point Estimate
     b) Interval Estimation
  6. Hypothesis Testing → Used to test claims or assumptions
  7. Confidence Interval (CI)
  8. Regression Analysis



(8) Regression Analysis

(9) Sampling Distribution

(10) Central Limit Theorem

(11) Types of Inferential Methods → a) Parametric Tests
b) Non-Parametric Tests.

* Point Estimation :-

Estimating population mean using sample mean.

Commonly Used Point Estimates are :-

(1) Sample Mean ($\bar{x}$)

(2) Sample std (σ)

(3) Sample Proportion (p)

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Sample size :-

- $n \leq 30$ ⇝ fluctuate & unreliable estimates
- $n > 100$ ⇝ stable estimates
- $n \rightarrow$ very large ⇝ better represent the population.

where, if population length = 10000
then, sample size (n) = 50

* Sampling Techniques :-

(1) Convenience Sampling ⇝ sample is selected based on its availability.

(2) Systematic Sampling ⇝ sample is selected at regular intervals

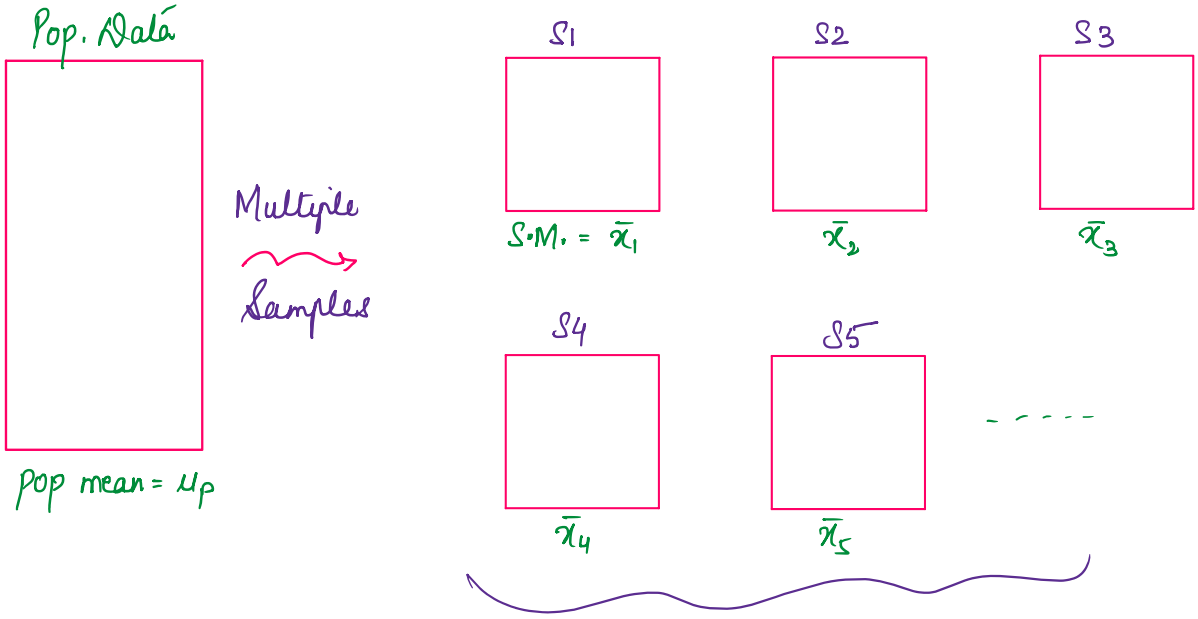
(3) Simple Random Sampling ⇝ sample is selected when it has equal chance of being selected

(4) [illegible] Sampling ⇝ [illegible]

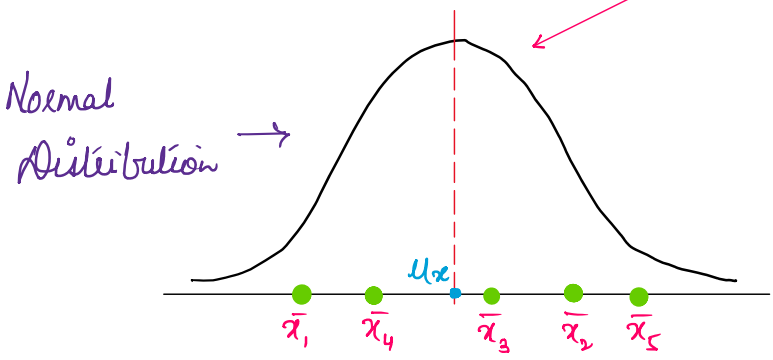
equal chance of being selected

④ Stratified Sampling ⇝ sample is created by dividing the pop. into different groups.

* Central Limit Theorem:-



Draw a distribution of samples means



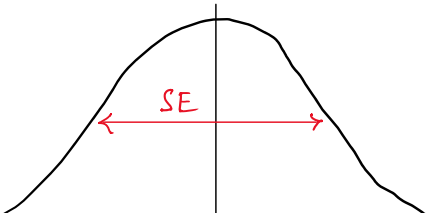
$\mu_{\bar{x}}$ = Mean of sampling distribution

& then,

$$\mu_{pop} \approx \mu_{\bar{x}}$$

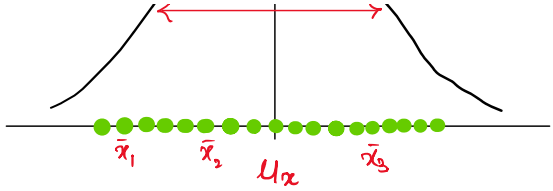
This is Central Limit Theorem.







Assume Original Dist.



Sampling Distribution from samples Means

- CLT states that when large no. of samples are drawn from a population, the sampling distribution of sample's means follows a normal distribution.
- Mean of sampling distribution ≈ Population Mean

$$\mu_{\bar{x}} \approx \mu_p$$

- The spread (std) of sampling distribution is called as standard error.

$$SE = \frac{\sigma}{\sqrt{n}}$$

where, σ = pop. std. dev.
n = sample size

- With increase in sample size, the standard error decreases.

Conditions for CLT:-

- Sample size should have to be → $n \geq 30$
- Samples must be independent of one another
- Samples must be randomly selected

★ Sampling Dist. for Proportions:-

It is a prob. dist. created from a large no. of samples drawn from a population

It is a prob. dist. created from a large ... of ... from a population

The standard dev. of sampling dist. is given by -

$$\sigma = \sqrt{\frac{P \cdot (1-P)}{n}}$$

} can be used when pop. std. dev is not known

In such cases,

Conditions for CLT :-

$n \cdot p \geq 10$ & $n \cdot (1-p) \geq 10$

where, p = population proportions

For point Estimation :-

It uses multiple samples built on CLT to create a robust methodology for estimating population parameters.

Standard Dev.	Standard Error
Measures the variability of individual observations around Mean	Measures the variability of a sample statistic $SE = \frac{\sigma}{\sqrt{n}}$

* Confidence Interval Estimation :-

Confidence Interval → Range of values within which there is a possibility of true population parameter

Instead of saying → The avg. height of the people of India is exactly 174 cm

is exactly 174 cm

We say → We are 90% confident that the avg. height of the people of India lie b/w 170 - 175 cm

∴ This range → 170 cm - 175 cm → confidence interval